

demo_commands

build your own packages

```
cd /usr/src
git clone https://github.com/freebsd/freebsd-src.git /usr/src
git checkout release/15.0.0
```

Build the packages by compiling from source, using make packages to create base system packages.

```
make -j8 buildkernel && make -j8 buildworld && make -j8 packages
```

checkout the next release

```
git checkout release/15.0.0-p1
```

and use update-packages for that

```
make -j8 buildkernel && make -j8 buildworld && make -j8 update-packages
```

repeat for every release until the last recent.

After building, the packages will get saved into

```
/usr/obj/usr/src/repo/FreeBSD:15:amd64/15.1px
```

use nginx to publish the contents of this folder:

```
vi /usr/local/etc/nginx/nginx.conf
```

```
server {
    listen      80;
    server_name builder;
    location / {
        alias /usr/obj/usr/src/repo/;
        autoindex on;
    }
}
```

Converting existing Hosts

Show freebsd-version

```
freebsd-version -kru
```

Run pkgbasify script:

```
./pkgbasify.lua
```

confirm conversion and creation of boot environment

wait for it to finish

reboot

```
shutdown -r now
```

Patch level update

show version after pkgbasify:

```
freebsd-version -kru
```

create boot environment

```
bectl create 150p10  
mkdir /mnt/upgrade  
bectl mount 150p10 /mnt/upgrade
```

go to edit the configuration file (this is only needed for this demonstration purposes).

```
vi /mnt/upgrade/usr/local/etc/pkg/repos/FreeBSD-base.conf
```

change 15.0 to base_release_0

```
FreeBSD-base: {  
  url: "http://builder/${ABI}/base_release_0",  
  enabled: yes  
}
```

```
pkg -c /mnt/upgrade upgrade
```

Time:

5.30 real

0.99 user

3.23 sys

activate boot env

```
bectl activate -t 150p10
```

reboot

```
shutdown -r now
```

```
bectl activate 150p10
```

minor upgrade

show version

```
freebsd-version -kru
```

create boot environment

```
bectl create 151  
bectl mount 151 /mnt/upgrade/
```

change config to base_release_1

```
vi /mnt/upgrade/usr/local/etc/pkg/repos/FreeBSD-base.conf
```

```
FreeBSD-base: {  
  url: "http://builder/${ABI}/base_release_1",  
  enabled: yes  
}
```

update boot environment

```
pkg -c /mnt/upgrade upgrade
```

Time:

6.43 real

1.74 user

3.24 sys

temporarily activate it and restart

```
bectl activate -t 151  
shutdown -r now
```

make it permanent

```
bectl activate 151
```

major upgrade

using bectl:

```
bectl create 16  
bectl mount 16 /mnt/upgrade
```

change config to base_latest

```
vi /mnt/upgrade/usr/local/etc/pkg/repos/FreeBSD-base.conf
```

```
FreeBSD-base: {  
  url: "http://builder/${ABI}/base_latest",  
  enabled: yes  
}
```

upgrade

```
env ABI=FreeBSD:16:amd64 pkg -c /mnt/upgrade upgrade -r FreeBSD-base  
env ABI=FreeBSD:16:amd64 pkg -c /mnt/upgrade upgrade -r FreeBSD-ports
```

Time (base):

6.35 real

2.34 user

2.83 sys

activate boot environment and restart

```
bectl activate -t 16  
shutdown -r now
```

activate be permanently

```
bectl activate 16
```

major upgrade 14-15

using bectl:

```
bectl create 15  
bectl mount 15 /mnt/upgrade
```

change config to base_latest

```
vi /mnt/upgrade/usr/local/etc/pkg/repos/FreeBSD-base.conf
```

```
FreeBSD-base: {  
  url: "http://builder/${ABI}/base_release_0",  
  enabled: yes  
}
```

upgrade

```
env ABI=FreeBSD:15:amd64 pkg -c /mnt/upgrade install FreeBSD-set-minimal  
FreeBSD-kernel-generic FreeBSD-ssh FreeBSD-bsdconfig  
pkg -c /mnt/upgrade remove -fy -g 'FreeBSD-*14.3*'  
env ABI=FreeBSD:15:amd64 pkg -c /mnt/upgrade install -f FreeBSD-set-minimal  
FreeBSD-kernel-generic FreeBSD-ssh FreeBSD-bsdconfig
```

change ports and kmods config

```
cp /etc/pkg/FreeBSD.conf /mnt/upgrade/usr/local/etc/pkg/repos/  
vi /mnt/upgrade/usr/local/etc/pkg/repos/FreeBSD.conf
```

FreeBSD-ports, FreeBSD-ports-kmods (add ports in repo name)

```
env ABI=FreeBSD:15:amd64 pkg -c /mnt/upgrade upgrade -r FreeBSD-ports
```

activate boot environment and restart

```
bectl activate -t 15  
shutdown -r now
```

activate be permanently

```
bectl activate 15
```

Jail with base system packages

create dataset

```
zfs create tank/foo
```

if packages are not signed, you can simply use the pkg rootdir option

```
pkg -r /tank/foo/ install FreeBSD-set-minimal-jail
```

enable jails on the host

```
sysrc jail_enable=YES
```

install FreeBSD-jail and FreeBSD-bsdconfig if using a minimal host

```
pkg install FreeBSD-jail FreeBSD-bsdconfig
```

else use bsinstall like documented in jail(8) EXAMPLES

```
bsinstall jail /tank/foo/
```

if you are using your own pkg configuration, give the BSDINSTALL_PKG_REPOS_DIR

```
env BSDINSTALL_PKG_REPOS_DIR=/usr/local/etc/pkg/repos/ bsinstall jail  
/tank/foo/
```

create a pkg conf

```
vim /etc/jail.conf.d/pkgbase.conf
```

```
pkgbase {  
    path = /zroot/jails/pkgbase;  
    mount.devfs;  
    host.hostname = pkgbase;  
    ip4 = inherit;  
    exec.start = "/bin/sh /etc/rc";  
    exec.stop = "/bin/sh /etc/rc.shutdown jail";  
}
```

copy resolv conf inside the jail

```
cp /etc/resolv.conf /zroot/jails/pkgbase/etc/
```

start and enter the jail

```
service jail start pkgbase  
jexec pkgbase
```

Poudriere pkgbase jail

fetch via url

```
poudriere jail -c -j 16current -m pkgbase=latest -v 16 -U http://pkgbase-  
host.local
```

Build from local source tree

```
poudriere jail -c -B -j 16current -m src=/usr/src -b -v 16-CURRENT -K GENERIC
```